MILITARY SPECIFICATION

SCREEN, X-RAY PROTECTIVE, Frame Mounted, Mobile

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of frame mounted mobile radiation protective device suitable for use in protecting a patient from scattered radiation during chest, skull and spine radiography or by draping over table for extremities.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-C-82	Casters, Swivel, Chair
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-T-830a	Tube, Steel (Carbon, Mechanical; Seamless and Welded)
PPP-B-636	Box, Fiberboard

STANDARDS

Federal

FED-STD-751	Stitches, Seams and Stitchings

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6525

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Fire Protection Association (NFPA)
56 Code for the Use of Flammable Anesthetics

Application for copies should be addressed to
National Fire Protection Association
60 Batterymarch Street, Boston, Mass. 62110

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.1), prior to the start of production of the contract quantity, one preproduction sample of the X-ray protective screen shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted, together with the contractor's inspection and test reports, to the contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the preproduction sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Material

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3.2.1 Screen. Screen shall be a single or multiple ply flexible vinyl plastic sheet, which shall be uniformly impregnated with lead or lead compound equivalent to not less than 0.25 millimeter (mm.) of lead.

3.2.2 Tubing. Tubing used in the manufacture of the screen shall be 1010 mechanical carbon tube steel conforming to QQ-T-830a.

3.2.3 Casters. Casters shall be type I, style a, size B, conforming to FF-C-82a.

3.2.4 Stitching thread. Thread shall be nylon thread or 10/4 cotton thread having a minimum breaking strength of 8 pounds

3.2.5 Tape. Tape shall be cotton or plastic.

3.2.6 Welding rod. Welding rod used shall have the same material composition as the welded members.

3.3 Construction. The X-ray protective screen shall consist of a radiopaque screen supported by a mobile, caster mounted frame . The frame shall consist of a base section and a screen support, and shall be constructed in accordance with figure 1. The screen support shall telescope freely into the base section and shall have a minimum adjustable height range of 33-1/4 inches to 47-3/4 inches, as measured from the floor level. The screen support height shall be held in position by means of thumbscrews.

3.3 1 Base section. The base section components shall be fabricated of tubing 1-inch O.D., 0.049 inch (No. 18, Birmingham Wire (B.W.) gage) thick. The base section shall consist of two uprights, two horizontal members and two cross braces, having the following lengths:

Horizontal member - 18 inches ± 1/4-inch
Cross brace - 25 inches ± 1/4-inch

Each horizontal member shall have two holes drilled 4 inches to 6-1/8 inches apart to pass bolts to secure the assembled cross braces to the horizontal members. Each end of the cross brace shall be cut to fit against the horizontal member without excessive crevices. A nut shall be brazed or welded in the open ends of the cross brace to receive the bolt. Two additional holes shall be drilled through each horizontal member, as shown in figure 1, to accommodate caster supports. The distance between centers of the casters

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shall be at least 15-5/16 inches. The open ends of the horizontal tubing shall be closed with close-fitting hole plugs. Each upright shall be joined to the center of the horizontal member by continuous welding, as shown in figure 1. A spring shall be fitted inside each vertical member to prevent the telescoping screen support from striking the bottom. The top of the vertical member shall be reinforced by a minimum 3/16-inch thick collar, through which shall pass a thumbscrew for engaging the telescoping screen support.

3.3.2 Screen support. The screen support shall consist of two uprights and a cross member joined together by continuous welding, as shown in figure 1. The screen support components shall be fabricated of tubing 7/8-inch O.D., 0.049 inch (No. 18, Birmingham Wire (B.W.) gage) thick. The ends of the crossmember shall be cut to fit against the uprights without excessive crevices. The crossmember shall be located 1 inch $\pm$ 1/8 inch from the top of the uprights. The bottom of each upright shall be provided with a stop to prevent the screen support's removal from the base section when it is fully extended. The top of the upright tubing shall be closed with close-fitting hole plugs. The screen support shall be constructed so as to move freely within the base section without binding.

3.3.3. Screen. A radiopaque screen shall be supplied which shall fold over the cross member of the screen support, as shown in figure 1. The overall size of the screen shall be 96 inches long by 24 inches wide with a tolerance of $\pm$ 1/4 inches on both dimensions. Both sides of the screen shall have a pair of holes which shall be reinforced by metal grommets. The holes shall be symmetrically located about the screen's center length. To prevent the grommets from tearing out, each grommet shall be reinforced with a cloth which shall be neatly trimmed around the grommet. The sides and ends of the screen shall be bound with tape and stitched. The stitches that are applied shall be type 301, six to eight per inch, conforming to FED-STD-751. The ends of the screen shall be aligned, when the screen is folded over the screen support. The screen shall be secured to the screen support by means of two pairs of binding screws and binding posts which shall pass through the metal grommets in the screen. If the screen is layered, the binding shall be continuous and permanent so that no separation will occur.

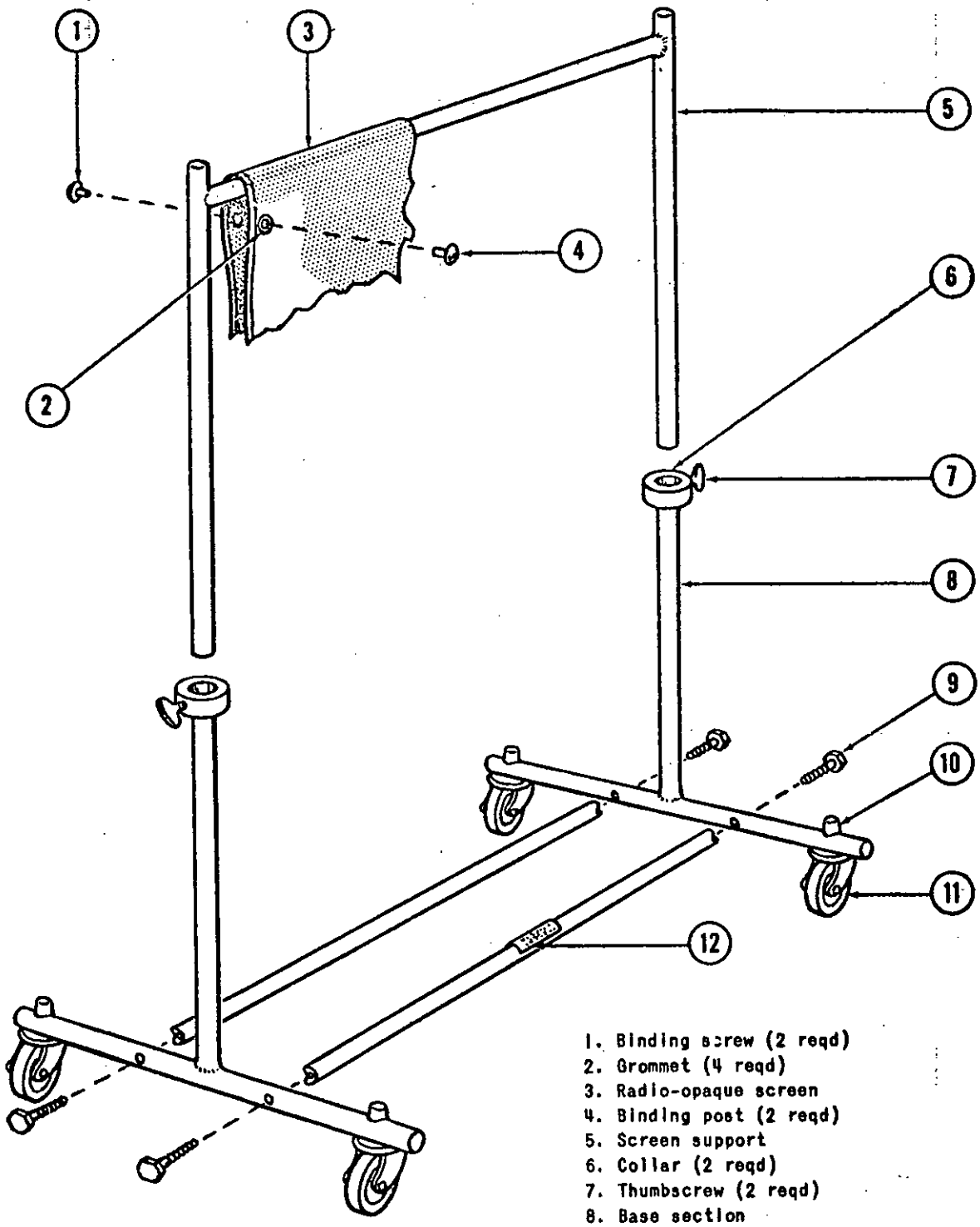
3.4 Components.

3.4.1 Screen. The lead impregnated plastic sheet shall have the following properties when tested per 4.4.1.3, 4.4.1.4, and 4.4.1.5:

Weight/square foot -pounds, max.	0.9
Tensile strength-minimum	
Original-pounds/square inch	1000
After aging - percent	75

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Figure 1



1. Binding screw (2 reqd)
2. Grommet (4 reqd)
3. Radio-opaque screen
4. Binding post (2 reqd)
5. Screen support
6. Collar (2 reqd)
7. Thumbscrew (2 reqd)
8. Base section
9. Bolt (4 reqd)
10. Caster support (4 reqd)
11. Electrically conductive caster (4 reqd)
12. Nameplate

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The lead impregnation shall provide a lead equivalency of not less than 0.25 millimeter (mm.), when tested per 4.4.1.1.1. There shall be no change or loss in radiation protective value, when tested per 4.4.1.2.

3.4.2 Casters. The casters that shall be supplied shall be electrically conductive. The resistance of the caster mounted frame shall not exceed 250,000 ohms when tested per 4.4.2.1.

3.5 Finish. Prior to the finishing application the welded frame shall be thoroughly cleaned to remove all grease, oil, dirt, rust, etc. The finish that shall be applied to the frame shall be chromium plated over nickel over copper. The chromium plating shall be as specified in QQ-C-320 for class 1, type 1. The nickel plating shall be as specified in QQ-N-290 for class 1, type III. In addition, hole plugs, thumbscrews, hardware, caster supports and metal parts of casters shall be suitably plated to prevent corrosion.

3.6 Identification marking.

3.6.1 Frame. A metal nameplate shall be securely attached to one of the lower cross braces. The nameplate shall bear the following information:

Contract or purchase order number

Federal stock number

Name or registered trademark of manufacturer

U.S.

Marking secured by adhesives shall not be used.

3.6.2 Screen. The surface of each screen shall be permanently and legibly marked with the following:

Name or registered trademark or manufacturer

Lead equivalency of screen

X-ray tube peak voltage to which lead equivalency applies.

3.7 Workmanship Workmanship shall be first class throughout. The screens and frames shall be free from defects which detract from their appearance or may impair their serviceability.

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3.7.1 Welding. Welds shall be thorough, nonporous; free from pits, cracks and other imperfections. All welds shall be cleaned, and all exposed welds shall be ground and finished smooth to match adjoining surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Military Standard MIL-STD-105.

	Inspection level	AQL (Defects per hundred)
For visual examination		
Major defects	II	1.0
Total (major and minor combined) defects	II	4.0
For dimensional examination	S-3	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with the following table:

Component	Characteristic	Requirement	Test Procedure	Inspection level
Screen	Distribution	3.4.1	4.4.1.1.1	(1)
	Radiation deterioration	3.4.1	4.4.1.2	(2)
	Tensile strength			(2)
	Before aging	3.4.1	4.4.1.3	
	After aging	3.4.1	4.4.1.4	
			4.4.1.3	
	Weight	3.4.1	4.4.1.5	(2)
Frame & Casters	Electrical conductivity	3.4.2	4.4.2.1	(3)

(1) Test for distribution shall be performed at three points in each of three locations on each roll of sheeting making a total of nine determinations per roll. Locations shall be as follows:

(a) Two feet from end of roll.
(b) Center of roll.
(c) Two feet from other end of roll.

At each of the above locations, the distribution shall be determined at each edge and at the midpoint. Any value falling below the value specified shall be considered cause for rejection of the entire lot.

(2) Two samples, each 1 square foot, shall be selected from each roll of lead-impregnated plastic sheeting. One sample shall be used for radiation deterioration, and the other sample shall be used to determine both the weight and the tensile strength before and after aging. The acceptance number shall be zero.

(3) Acceptance number shall be zero, and not less than three samples shall be taken regardless of lot sizes.

4.3 Examination. The screens and frames shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101. Screen not complete with all component parts.
102. Thumbscrew does not tighten screen support in position.
103. Screen support does not move freely throughout its full excursion.
104. Bolts for securing cross braces do not thread properly into bolt to form a neat and secure joint.
105. Collar at top of upright of base section not flush with top of tubing; not properly secured in position.
106. Plastic sheeting not free of cuts, tears, holes, perforations, and blisters.
107. Casters do not fit base.

Minor

201. Casters do not roll or swivel freely.
202. Hole plugs not securely in place or do not fit properly.
203. Ends of cross braces not cut to fit neatly against horizontal members of base section.
204. Stops on screen support do not prevent its removal.
205. Building posts and binding screws cannot be tightened and loosened.
206. Grommets not secure.
207. Hole of grommet not of proper size to prevent removal of assembled binding post.
208. Identification marking missing, incomplete, illegible, impermanent, and not securely attached.
209. Special marking on plastic screens missing, incomplete, illegible, and impermanent.
210. Welds not free from pits, cracks, and other imperfections.
211. Welds not ground and finished smooth to match adjoining surfaces; not free from undercutting.
212. Surfaces of plastic sheeting not smooth, uniform, and free from tack, lumps, imbedded foreign matter, and pits.
213. Plastic sheeting not clean and free from abrasions, surface dirt, stains, and other foreign matter.

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214. Stitches loose, poorly formed, crooked, and skipped.
215. Stitching not free from puckers, twists, runoffs, and exposed raw edges.
216. Spring in uprights of base section does not break fall of the screen support.
217. Binding not free from tears and frayed edges.
218. Plating not free from scratches, pits, rough areas, and other surface defects.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The screens shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Plastic sheeting.

4.4.1.1 X-ray opacity. Radiographic tests to determine compliance with specification requirements shall be conducted under the following exposure conditions: Cardboard holders and standard X-ray films shall be used. X-ray films shall be processed in accordance with the manufacturer's recommendations. All test exposures shall be carried out with 75 kilovolts-peak. Milliampere-seconds and distance shall be so adjusted that the density of the exposed film at the 0.25-mm. step of the lead ladder shall be 1.0 to 1.5.

4.4.1.1.1 Distribution test procedure. Samples of the plastic sheeting shall be placed alongside a lead penetrometer having steps from 0.1 mm. to not less than 1.0 mm. in penetration value, including a step at 0.25 mm. An X-ray exposure shall be made of both simultaneously. After development, a comparison of densities shall be made by means of a suitable densitometer.

4.4.1.2 Radiation deterioration test. Lead-impregnated vinyl plastic sheet shall be subjected to massive X-ray radiation in the range of 200,000 roentgens.

4.4.1.3 Tensile strength. Tests for tensile strength shall be conducted in accordance with method 4111, as specified in FED-STD-601, using dumbbell specimens (see 2.1.1 of method 4111).

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4.4.1.4 Accelerated aging test. Samples of the plastic sheeting shall be subjected to a temperature of 158° Fahrenheit (F.) ± 2°F. in air for 7 days.

4.4.1.5 Weight. The weight shall be determined by weighing a one-square foot sample, accurate to the nearest 0.01 pound.

4.4.2 Casters and frame.

4.4.2.1 Electrical conductivity. Tests for electrical conductivity shall be performed in accordance with section 2533 of National Fire Protection Association (NFPA) 56.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1)screen as specified constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. Each screen shall be disassembled for as far as practicable. The disassembled components shall be wrapped with kraft paper and secured in an exterior container in such a manner to insure against marring, scratching, shifting, or damage to the item.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Closure. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform to 30.4, of PPP-B-636.

5.2.1.3 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

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NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be a box of appropriate size and design and constructed in accordance with PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1.2 and 5.2.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

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CATEGORY II.

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